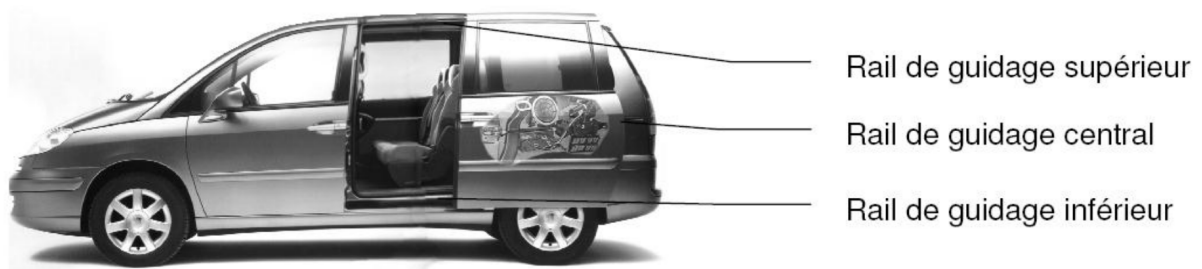


Minivan side door control

System Presentation

To face the increasing competition, automobile manufacturers are conceiving various accessories to satisfy an ever more demanding clientele. In this logic, some minivans have two sliding side doors (noted later PLC) equipped with an electromechanical system of automatic opening and closing. Each sliding door is guided by three rails attached to the vehicle body.



The studied system allows to drive the PLC at the level of the central guide rail, by means of a mobile carriage set, guided by a system of rollers and wrapped around a drum, whose rotation is obtained via a geared motor. The winding of the cables around the drum makes possible the door movement. The central computer of the vehicle and the various concerned components communicate via a CAN2.0B bus. The opening time of a PLC must be at most 3 seconds. In addition, the central computer must detect a PLC failure in a maximum duration of 3.6ms.

The objective of this study is to dimension the CAN bus (in terms of transmission rate) connecting the central computer and the various sensors / actuators, in order to meet the requirements of a PLC command.

N° Frame	Id	Length (bytes)	DATA
1	755	4	03 30 80 00
[2 :2 :34]	655	4	03 70 80 01
[3 :2 :35]	755	4	03 30 80 01
36	655	4	03 70 80 11

Tab1 : frames on CAN bus when opening the PLC

For a request (id 755) 4 bytes of hexadecimal description : • header : 03 30 • Command : • 80 : left PLC • A0 : steering left lock actuator • B0 : right PLC • D0 : steering right lock actuator • E0 : door movement sound emission • Command Type: ✓ 00 : open ✓ 01 : state ✓ 11 : stop	For a feedback (id 655) 4 bytes of hexadecimal description • Header : 03 70 • Commanded output : • 80 : Left PLC • A0 : Left Lock • B0 : Right PLC • D0 : Right Lock • E0 : buzzer • Command Type: ✓ 00 : failed ✓ 01 : under processing ✓ 11 : finished
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Tab2 : Frames description

CAN bus Dimensioning

1. Explain the coding of the DATA field of each frame of tab1 found on the CAN bus.
2. What is the minimum data rate required on the CAN bus to ensure the maximum opening time of the door?
3. What is the minimum data rate required on the CAN bus to ensure maximum fault detection time by the central computer?
4. Deduce the minimum rate of the CAN bus that meets both constraints.